

Minimum Lap Time Optimization for a Double-Track Vehicle Model

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Abstract—This report presents an optimal control framework for minimum lap time optimization of a race vehicle modeled using a nonlinear double-track formulation on two circuits: the Berlin Formula E track and the Sochi Autodrom. The continuous-time optimal control problem (OCP) incorporates detailed vehicle dynamics, including nonlinear Pacejka tire forces, aerodynamic effects, load transfer, and combined-slip limitations enforced through Kamm’s friction circle constraints. Additional constraints include actuator limits, rate bounds, and track boundary conditions defined in spatial coordinates.

The problem is formulated along the track arc length, allowing time to be reconstructed implicitly. The resulting OCP is transcribed into a finite-dimensional nonlinear program (NLP) using direct collocation with trapezoidal integration on a uniform spatial grid. The resulting large-scale nonlinear program is solved using the AMIGO framework with IPOPT, leveraging automatic differentiation for efficient and robust convergence.

Track data is preprocessed from raw CSV and GeoJSON formats into a smooth arc-length parameterization. Curvature profiles are computed and regularized using either Gaussian filtering or spline-based smoothing to ensure numerical stability and physical consistency. For the Sochi circuit, geographic coordinates are converted into a local Cartesian frame, and periodic smoothing is applied to maintain continuity.

The optimization produces optimal racing lines, velocity profiles, and control inputs that fully exploit tire friction and available power while satisfying all constraints. The Berlin track results in a shorter lap time of 86.80 s, while the Sochi track yields 192.50 s, reflecting differences in track length and geometry. Key challenges include sensitivity to initialization, scaling of variables, and constraint formulation, all of which are addressed through systematic tuning.

I. OPTIMAL CONTROL PROBLEM FORMULATION

The minimum-lap-time problem is formulated as a nonlinear OCP in spatial coordinates $s \in [0, s_f]$, where time is implicitly reconstructed along the trajectory. The system is transcribed into a finite-dimensional NLP using direct collocation.

1) *State and Control Variables*: The state vector is

$$x = [V \quad \beta \quad \omega_z \quad n \quad \xi]^\top, \quad (1)$$

representing velocity, sideslip, yaw rate, lateral displacement, and heading angle.

The control vector is

$$u = [\delta \quad F_{\text{drive}} \quad F_{\text{brake}} \quad \Gamma_y]^\top, \quad (2)$$

i.e., steering angle, total driving and braking force, and lateral load transfer force.

2) *Objective*: The objective is to minimize lap time:

$$\min T \quad (3)$$

with

$$T = \int_0^{s_f} \frac{1 - n\kappa}{V \cos(\xi + \beta)} ds, \quad (4)$$

plus a small regularization term penalizing control rate changes:

$$J_{\text{reg}} = r_\delta (\Delta\delta)^2 + r_F (\Delta F_{\text{net}})^2. \quad (5)$$

3) *Dynamics*: The spatial dynamics are written as

$$\frac{dx}{ds} = f(x, u, \kappa), \quad (6)$$

with scaling factor

$$SF = \frac{1 - n\kappa}{V \cos(\xi + \beta)}. \quad (7)$$

The model equations are therefore:

$$\frac{dV}{ds} = \frac{F_x}{m} SF, \quad \frac{d\beta}{ds} = \left(-\omega_z + \frac{F_y}{mV} \right) SF, \quad (8)$$

$$\frac{d\omega_z}{ds} = \frac{M_z}{J_{zz}} SF, \quad \frac{dn}{ds} = V \sin(\xi + \beta) SF, \quad (9)$$

$$\frac{d\xi}{ds} = \omega_z SF - \kappa, \quad (10)$$

where tire forces F_x, F_y, M_z are computed via a nonlinear Pacejka model including load transfer and aerodynamic effects.

4) *Constraints and Discretization*: The system is subject to physical constraints on tire forces, steering, and velocities. The continuous OCP is discretized using direct collocation, yielding a finite-dimensional NLP in the state and control variables at collocation points, enforcing both dynamics and constraints at each segment.

A. *Constraints*

1) *Path Constraints*: At each discretization node, the following constraints are enforced:

a) *Friction Circle Constraint*:

$$\frac{F_x^2 + F_y^2}{(\mu F_z)^2} \leq 1 \quad (11)$$

b) *Engine Power Constraint*:

$$\frac{VF_{\text{drive}}}{P_{\text{max}}} \leq 1 \quad (12)$$

c) *Mutual Exclusion Constraint*:

$$F_{\text{drive}} \cdot F_{\text{brake}} \leq 1e^{-3} \quad (13)$$

Note that the constraint does not strictly enforce ≤ 0 since the solver struggles to converge otherwise.

d) *Lateral Load Transfer Consistency*:

$$\Gamma_y = \frac{h_{cog}}{t_w} (\text{lateral force terms}) \quad (14)$$

2) *State and Control Bounds*: The variables are bounded as follows:

- Velocity constraint: $0 < V \leq V_{\max}$
- Side-slip constraint: $|\beta| \leq \beta_{\max}$
- Yaw rate constraint: $|\omega_z| \leq \omega_{z,\max}$
- Steering constraint: $|\delta| \leq \delta_{\max}$
- Drive force constraint: $0 \leq F_{\text{drive}} \leq F_{\text{drive},\max}$
- Brake force constraint: $F_{\text{brake},\min} \leq F_{\text{brake}} \leq 0$

The bound values are listed in Section VII Table II.

The lateral position is constrained by the track geometry:

$$-w_{\text{right}}(s) \leq n(s) \leq w_{\text{left}}(s) \quad (15)$$

3) *Rate Constraints*: Actuator dynamics are limited by:

$$\left| \frac{d\delta}{dt} \right| \leq \frac{\delta_{\max}}{T_\delta} \quad (16)$$

$$\left| \frac{dF_{\text{drive}}}{dt} \right| \leq \frac{F_{\text{drive},\max}}{T_{\text{drive}}} \quad (17)$$

$$\left| \frac{dF_{\text{brake}}}{dt} \right| \leq \frac{|F_{\text{brake},\min}|}{T_{\text{brake}}} \quad (18)$$

4) *Boundary Conditions*: Since a full lap is considered, periodic boundary conditions are imposed:

$$x(0) = x(s_f), \quad u(0) = u(s_f) \quad (19)$$

B. Direct Collocation Transcription

The continuous optimal control problem is discretized using direct collocation with trapezoidal integration.

The track is divided into N intervals of size:

$$ds = \frac{s_f}{N} \quad (20)$$

The NLP decision variables are states at nodes: x_i and controls at nodes: u_i . For each interval i , the dynamics are enforced using the trapezoidal rule:

$$x_{i+1} - x_i - \frac{ds}{2} (f(x_i, u_i) + f(x_{i+1}, u_{i+1})) = 0 \quad (21)$$

Time is reconstructed as:

$$\Delta t_i = \frac{ds}{2} (SF_i + SF_{i+1}) \quad (22)$$

Thus, the total lap time becomes:

$$T \approx \sum_{i=0}^{N-1} \Delta t_i \quad (23)$$

The resulting nonlinear program is:

$$\min_{x_i, u_i} \sum_{i=0}^{N-1} \frac{ds}{2} (SF_i + SF_{i+1}) + J_{\text{reg}} \quad (24)$$

subject to:

- Collocation dynamics constraints
- Path constraints (friction, power, load transfer)
- State and control bounds

- Rate constraints
- Periodic boundary conditions

Therefore, the resulting formulation is a large-scale sparse nonlinear programming problem that captures full nonlinear vehicle dynamics, tire saturation effects, and track geometry constraints. It is solved efficiently using direct collocation, transforming the infinite-dimensional OCP into a finite-dimensional NLP.

II. TRACK DATA PROCESSING

This section describes the preprocessing pipeline used to convert raw track geometry (CSV/GeoJSON) into a smooth, differentiable representation for racing line optimization. Each track is represented as a sequence of 2D centerline points.

The processing steps are:

- 1) Load centerline data from CSV or GeoJSON.
- 2) Convert geographic coordinates to a local Cartesian frame (if required) using an equirectangular projection.
- 3) Remove duplicate or near-duplicate points for numerical stability.
- 4) Compute cumulative arc length:

$$s_i = \sum_{j=1}^i \sqrt{(x_j - x_{j-1})^2 + (y_j - y_{j-1})^2} \quad (25)$$

- 5) Compute curvature using finite differences:

$$\kappa(s) = \frac{x'(s)y''(s) - y'(s)x''(s)}{(x'(s)^2 + y'(s)^2)^{3/2}} \quad (26)$$

- 6) Smooth curvature to reduce numerical noise.
- 7) Resample all variables onto a uniform grid of N nodes.

A. Curvature Smoothing Approach

Raw curvature is sensitive to noise and discretization artifacts. Two smoothing methods are available.

1) *Gaussian Convolution Smoothing*: Curvature is filtered using a Gaussian kernel:

$$\kappa_{\text{smooth}}(s) = (\kappa * \mathcal{G}_\sigma)(s) \quad (27)$$

where $\mathcal{G}_\sigma$ is a normalized Gaussian with arc-length standard deviation σ . The implementation uses

$$\sigma_{\text{pts}} = \frac{\sigma_{\text{arc}}}{\Delta s}, \quad \sigma_{\text{arc}} = 3.0 \text{ m} \quad (28)$$

This corresponds to smoothing over approximately 5–15 points depending on resolution.

2) *Spline-Based Smoothing*: A variational formulation is also used:

$$\min_{\hat{\kappa}} \sum_i (\kappa_i - \hat{\kappa}(s_i))^2 + \lambda \int (\hat{\kappa}''(s))^2 ds \quad (29)$$

with effective smoothing strength controlled by $s = \sigma^2 N$. This preserves large-scale structure (corners and straights) while enforcing curvature continuity.

The smoothing parameter ($\sigma = 3.0$ m) balances noise reduction and feature preservation by removing high-frequency artifacts while maintaining sharp corner geometry and smooth second derivatives for optimization.

B. Track Geometry

Each track is represented as:

$$(x(s), y(s), \kappa(s), w_{\text{left}}(s), w_{\text{right}}(s)),$$

where x, y define the centerline, κ curvature, and $w_{\text{left/right}}$ track boundaries.

1) *Berlin 2018*: Berlin has length 2326.91 m with 100 intervals ($ds = 23.27$ m) and curvature range $[-0.0810, 0.0796]$ m^{-1} . Raw curvature contains high-frequency noise and spikes, which are removed via spline-based smoothing. The final centerline remains fully within track boundaries and preserves the original geometry (Fig. 1).

2) *Sochi 2014*: Sochi is longer (5832.61 m, 300 intervals, $ds = 19.44$ m) with curvature $[-0.0367, 0.0439]$ m^{-1} . Geometry is extracted from GeoJSON and converted to Cartesian coordinates. Raw data contains non-physical oscillations due to discontinuities.

These are removed using cubic spline filtering in arc-length space and periodic curvature smoothing, with boundary wrapping to ensure continuity of derivatives. Compared to Berlin, stronger smoothing is required due to higher data variability. The final centerline remains fully contained within track limits and preserves global race-line structure (Fig. 2).

III. VEHICLE MODEL VERIFICATION

The vehicle is modeled as a double-track system with nonlinear tire forces. Refer to Section I for an overview of the state and input vectors and to Section VII for an overview of the parameters. The lateral tire forces are computed using a nonlinear tire model (Pacejka "Magic Formula"), which captures the relationship between slip angle and generated force:

$$F_{y,ij} = \mu_{ij} F_{z,ij} \left(1 + \epsilon_i \frac{F_{z,ij}}{F_{z0,i}} \right) \Phi(\alpha_{ij}) \quad (30)$$

$$\Phi(\alpha) = \sin(C_i \arctan(B_i \alpha - E_i (B_i \alpha - \arctan(B_i \alpha)))) \quad (31)$$

where:

- B is the stiffness factor,
- C is the shape factor,
- E is the curvature factor.

The front and rear axles use different parameterizations to reflect load transfer and geometry differences:

- Front tires: higher cornering stiffness due to steering compliance and load distribution.
- Rear tires: slightly reduced peak force but higher stability margins.

A. Kamm's Friction Circle Constraint

To ensure physical feasibility, combined longitudinal and lateral tire forces are constrained by the friction limit:

$$F_x^2 + F_y^2 \leq (\mu F_z)^2 \quad (32)$$

This constraint is enforced at all four wheels in the optimal solution and ensured by the code setup and optional diagnostic print statements.

The optimal solution respects Kamm's constraint across all wheels, with peak utilization occurring primarily at front outer wheel during corner entry (combined braking + steering load) and rear outer wheel during corner exit (combined traction + lateral load).

No violations of the friction boundary are observed, confirming that the optimizer remains within physically feasible limits.

IV. OPTIMIZATION RESULTS

Two optimal control problems were solved using the AMIGO interior-point solver for the Berlin 2018 circuit and the Sochi Autodrom 2014. The Berlin case converged in 106 iterations, while Sochi required 138 iterations, reflecting higher problem complexity.

A. Convergence and Computational Effort

Berlin converged faster in both iterations and runtime: 106 iterations and 86.80 s lap time over 2326.91 m, compared to Sochi with 138 iterations, 192.50 s, and 5832.61 m. The increased computational cost for Sochi is expected due to its longer track and additional complexity from GeoJSON-based preprocessing. This scaling is also reflected in the objective values (8.95×10^1 for Berlin and 1.95×10^2 for Sochi), consistent with the longer trajectory.

Both solutions operate near physical limits, with tire force constraints saturated at 1.0 and power usage around 0.93. This indicates that both trajectories fully exploit available friction and engine power throughout the lap.

B. Velocity

The velocity profiles, visualized in Figures 4, 7, 8 and 11, differ significantly: Berlin spans $[12.03, 42.50]$ m/s, while Sochi maintains a higher minimum speed of 14.62 m/s with the same upper bound. This suggests a smoother and more continuous driving line for Sochi, with fewer extreme deceleration phases.

Lateral motion and steering behaviour follow the same trend. Sochi stays slightly closer to the track center ($[-5.00, 5.00]$ m vs. $[-5.41, 5.33]$ m) and requires smaller steering inputs ($[-6.59^\circ, 5.36^\circ]$ vs. $[-9.07^\circ, 6.22^\circ]$), consistent with smoother curvature and reduced corner aggressiveness.

C. Periodicity and Dynamics

Both solutions satisfy periodic boundary conditions. However, the operating regimes differ. Berlin is a more aggressive solution with larger speed variations, lower slip angles, and reduced yaw activity (cf. Figure 4), while Sochi operates at higher sustained slip angles (β) and yaw rates (ω_z) (cf. Figure 8), indicating stronger continuous lateral loading, consistent with the tire force characteristics shown in the Pacejka plots in Figure 3.

Overall, Berlin produces a faster and more aggressive lap with lower computational effort, while Sochi yields a longer but smoother trajectory requiring more iterations and exhibiting higher dynamic load. Both solutions converge reliably and satisfy all physical constraints.

Across both tracks, the optimizer consistently prioritizes a trade-off between corner exit speed, early braking with late apexing, and full exploitation of tire friction without violation.

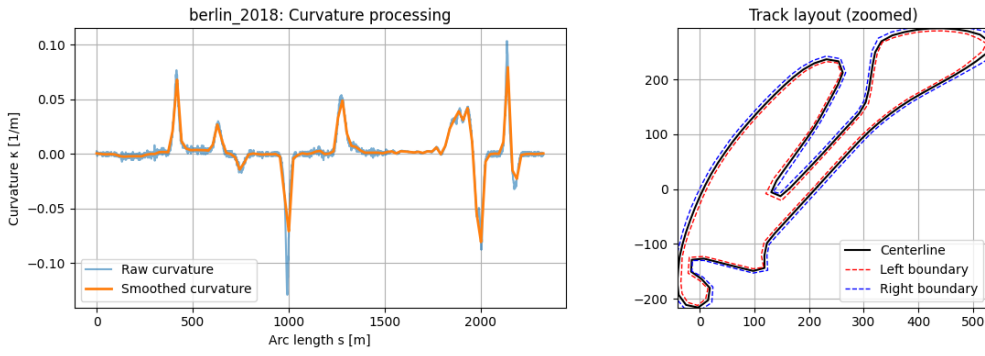


Fig. 1: Berlin 2018: raw vs smoothed curvature with centerline and boundaries.

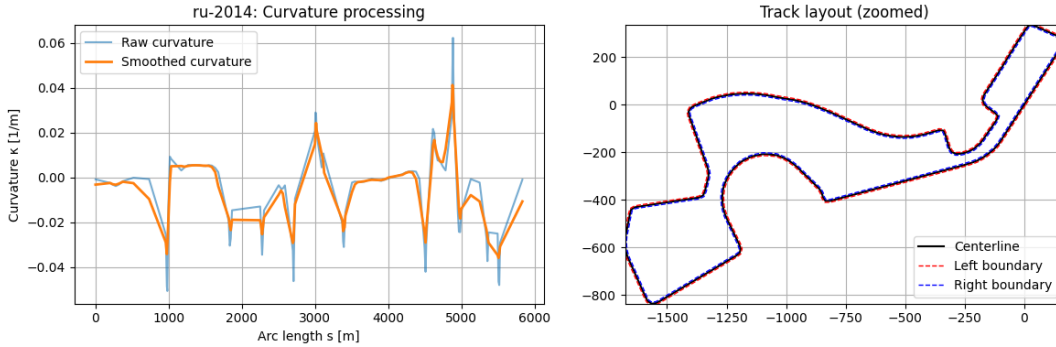


Fig. 2: Sochi 2014: raw vs smoothed curvature with centerline and boundaries.

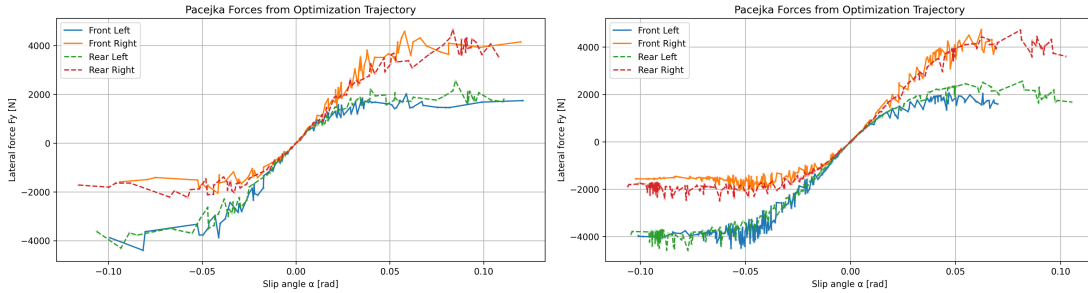


Fig. 3: Pacejka tire force characteristics for the optimized trajectories: Berlin 2018 (left) and Sochi 2014 (right), showing the nonlinear lateral force response under varying slip angles and load conditions.

V. CHALLENGES AND MODIFICATIONS

Several issues appeared during the development of the trajectory optimisation framework, mostly related to numerical stability, model sensitivity, and solver robustness. While the overall formulation worked as intended, a number of adjustments were necessary before reliable convergence was achieved.

- **Vehicle dynamics implementation:** Early versions of the model produced inconsistent results due to small errors in the dynamics formulation. To avoid spending excessive time debugging higher-level components, a simplified version of the model was first implemented and progressively extended. This stepwise approach made

it easier to isolate and correct modelling issues.

- **Scaling of variables:** Poor scaling initially led to unstable solver behaviour, including divergence in early runs. This was addressed by analysing typical magnitudes of all state and control variables and rescaling them to be approximately order one. Once consistent scaling was introduced, convergence became significantly more reliable. The final scaling choices are provided in Section VII Table III.
- **Initialisation strategy:** The solver proved sensitive to the initial guess, particularly for velocity, steering, and yaw-rate states. Several iterations were required to obtain a feasible starting trajectory that did not immediately lead to divergence. The final initialisation was constructed using physically consistent constant-speed assumptions, refined

through trial and error based on convergence behaviour. The final values are listed in Section VII Table II.

- **Constraint formulation:** Careful tuning of constraints was necessary to ensure numerical stability. In particular, both positive and negative bounds were required for drive and brake rate constraints to avoid artificial bias in actuator behaviour. Additionally, enforcing the mutual exclusion constraint too strictly (forcing it exactly to zero) led to convergence difficulties. Relaxing this constraint improved solver robustness without affecting physical plausibility.
- **Solver convergence behaviour:** In early experiments, the solver showed slow or oscillatory convergence when curvature variations were high. This was mitigated indirectly through improved scaling and better initial guesses rather than by modifying solver settings directly. After these adjustments, convergence became consistent across both track cases.

Overall, the most critical improvements came from consistent scaling and a physically meaningful initialisation strategy. Once these were in place, the optimisation framework became stable across different track geometries. Apart from these debugging-related adjustments and post-processing improvements, the implementation follows the assignment specification [1].

VI. COMPARATIVE ANALYSIS

This section compares the optimization results obtained on the Berlin 2018 racing track and the Sochi 2014 Autodrom. The two tracks differ significantly in length, curvature distribution, and corner sequencing, which directly influences the optimal driving strategy computed by the solver.

A. Track Characteristics and Driving Strategy

The Berlin 2018 track is shorter and features a higher proportion of short straights connected by medium-radius corners. This structure allows frequent acceleration phases and rewards a more aggressive longitudinal driving strategy. In contrast, the Sochi 2014 Autodrom is considerably longer and contains extended high-speed sections combined with several long, sweeping corners. This leads to a more continuous high-speed driving profile with fewer full acceleration-braking cycles.

These structural differences strongly influence the optimal trajectories. On the Berlin track, the solver prioritizes rapid speed changes and sharper steering inputs to maximize performance through tightly spaced corners. On the Sochi track, the optimal strategy shifts toward smoother control actions, maintaining higher average speed stability through long curvature transitions and minimizing unnecessary longitudinal losses.

B. Lap Time and Speed Performance

The resulting performance metrics reflect these differences in track layout:

- **Berlin 2018:** lap time = 86.80 s, speed range = [12.03, 42.50] m/s
- **Sochi 2014:** lap time = 192.50 s, speed range = [14.62, 42.50] m/s

Although both solutions reach the same maximum velocity, the Sochi solution maintains a higher minimum speed. This

indicates fewer full deceleration phases, consistent with the smoother curvature distribution and longer corner radii of the Sochi circuit. The Berlin solution, in contrast, exhibits stronger speed variation due to frequent transitions between straight segments and tighter corners.

C. Cornering Behavior and Control Effort

The steering and lateral dynamics further highlight the differences in driving strategy:

- **Berlin 2018 steering range:** $[-9.07^\circ, 6.22^\circ]$, lateral offset $[-5.41, 5.33]$ m
- **Sochi 2014 steering range:** $[-6.59^\circ, 5.36^\circ]$, lateral offset $[-5.00, 5.00]$ m

The Berlin solution requires larger steering inputs, reflecting tighter corner radii and more frequent directional changes. The Sochi solution exhibits reduced steering amplitudes, consistent with smoother curvature transitions. However, this comes with increased dynamic loading in the form of higher sustained slip angles and yaw rate activity, indicating that vehicle stability is maintained under more continuous lateral acceleration.

D. Solver Performance and Convergence

Both optimization runs achieved convergence to similar optimality tolerances, but with different computational effort:

- **Berlin 2018:** 106 iterations, 86.80 s computation time
- **Sochi 2014:** 138 iterations, 192.50 s computation time

The increased iteration count and runtime for the Sochi track can be attributed to its longer track and more complex curvature profile, which increases the dimensionality and difficulty of the nonlinear optimization problem. Despite this, both cases exhibit stable convergence behavior with smooth reduction in primal and dual infeasibilities.

VII. APPENDIX

REFERENCES

- [1] G. Kennedy, "2026 class project: Race car minimum lap time," *IEEE Transactions on Something*, 2026. [Online]. Available: https://smdogroup.github.io/ae6310/jupyter/2026_project/race_car_problem.html#Actuator-Rate-Constraints.

TABLE I: Optimizer Settings (Berlin and Sochi)

Parameter	Value
Initial barrier parameter	1×10^{-1}
Maximum iterations	300
Max. line search iterations	30
Convergence tolerance	1×10^{-7}
Init. least-squares multipliers	False
Barrier strategy	quality_function
Second-order correction	True
Filter line search	True
Verbose barrier output	False

TABLE II: Vehicle Model Parameters

Symbol	Value	Unit / Description
m	1200.0	kg (vehicle mass)
g	9.81	m/s ² (gravitational acceleration)
l_f	1.5	m (CG to front axle)
l_r	1.4	m (CG to rear axle)
t_{wf}	1.6	m (front track width)
t_{wr}	1.5	m (rear track width)
b_{veh}	2.0	m (vehicle width)
h_{cog}	0.4	m (CG height)
J_{zz}	1260.0	kg m ² (yaw inertia)
A	1.0	m ² (frontal area)
c_d	1.4	drag coefficient
c_{Lf}	2.4	front downforce coefficient
c_{Lr}	3.0	rear downforce coefficient
f_r	0.010	rolling resistance coefficient
ρ	1.2041	kg/m ³ (air density)
P_{max}	270000	W (max engine power)
$F_{drive,max}$	7100	N (max driving force)
$F_{brake,min}$	-20000	N (max braking force)
δ_{max}	0.4	rad (max steering angle)
v_{max}	42.5	m/s (max velocity)
k_{drive}	0.0	drive force distribution
k_{brake}	0.7	brake force distribution
k_{roll}	0.5	roll moment distribution
T_δ	0.2	steering time constant
T_{drive}	0.3	drive time constant
T_{brake}	0.2	brake time constant
B_f	9.62	Pacejka stiffness (front)
B_r	8.62	Pacejka stiffness (rear)
C_f	2.59	Pacejka shape (front)
C_r	2.65	Pacejka shape (rear)
E	1.0	curvature factor
F_{z0}	3000	N (nominal vertical load)
ϵ_{front}	-0.0813	load sensitivity (front)
ϵ_{rear}	-0.1263	load sensitivity (rear)
μ	1.0	friction coefficient

TABLE III: Scaling Factors

Variable	Scale	Description
t	60.0	lap time scale [s]
n	5.0	lateral position scale [m]
V	40.0	velocity scale [m/s]
ξ	0.15	heading angle scale [rad]
β	0.05	sideslip angle scale [rad]
ω_z	0.5	yaw rate scale [rad/s]
δ	0.1	steering angle scale [rad]
F_{drive}	7000.0	driving force scale [N]
F_{brake}	20000.0	braking force scale [N]
Γ_y	5000.0	lateral load transfer scale [N]

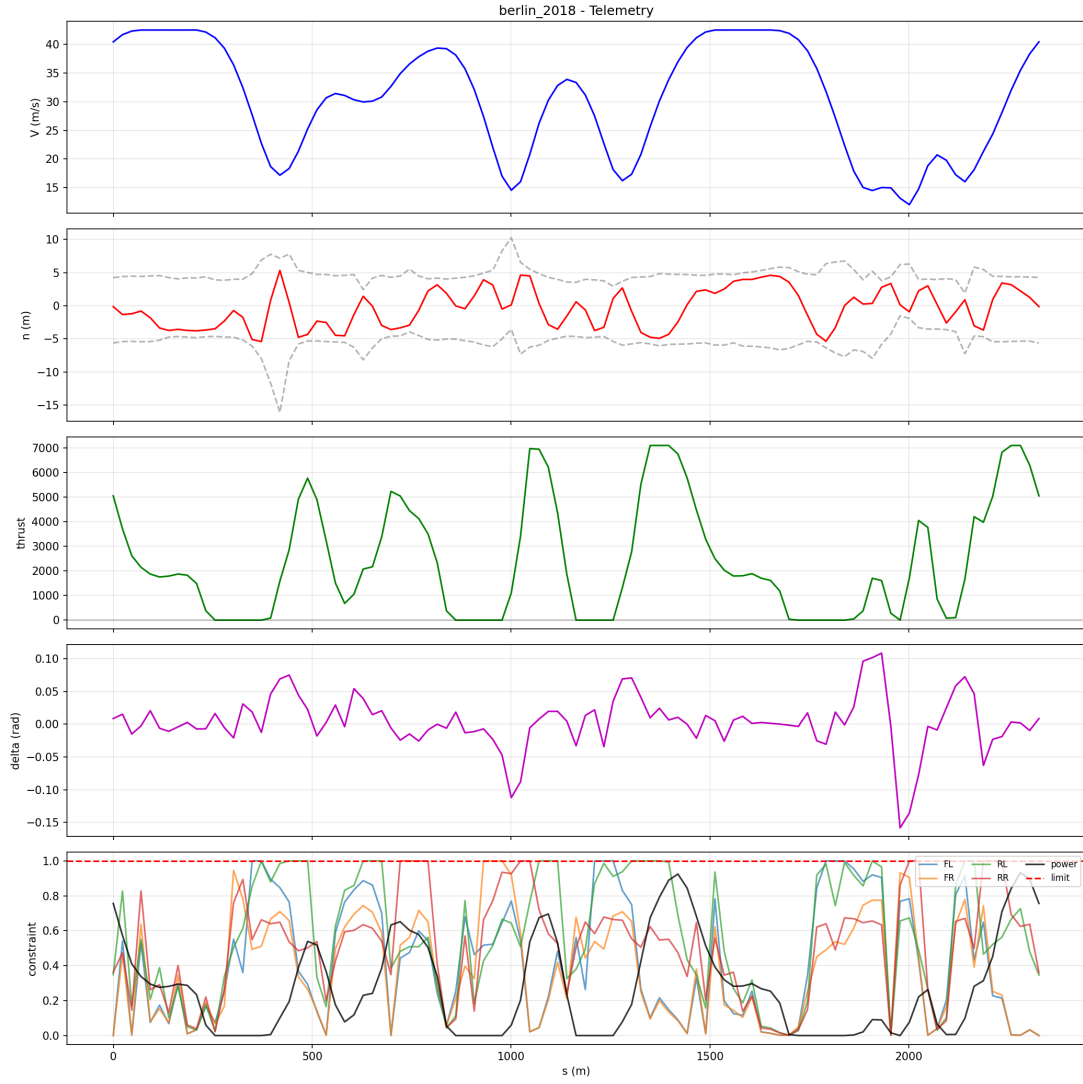


Fig. 4: Berlin 2018: vehicle telemetry over the lap.

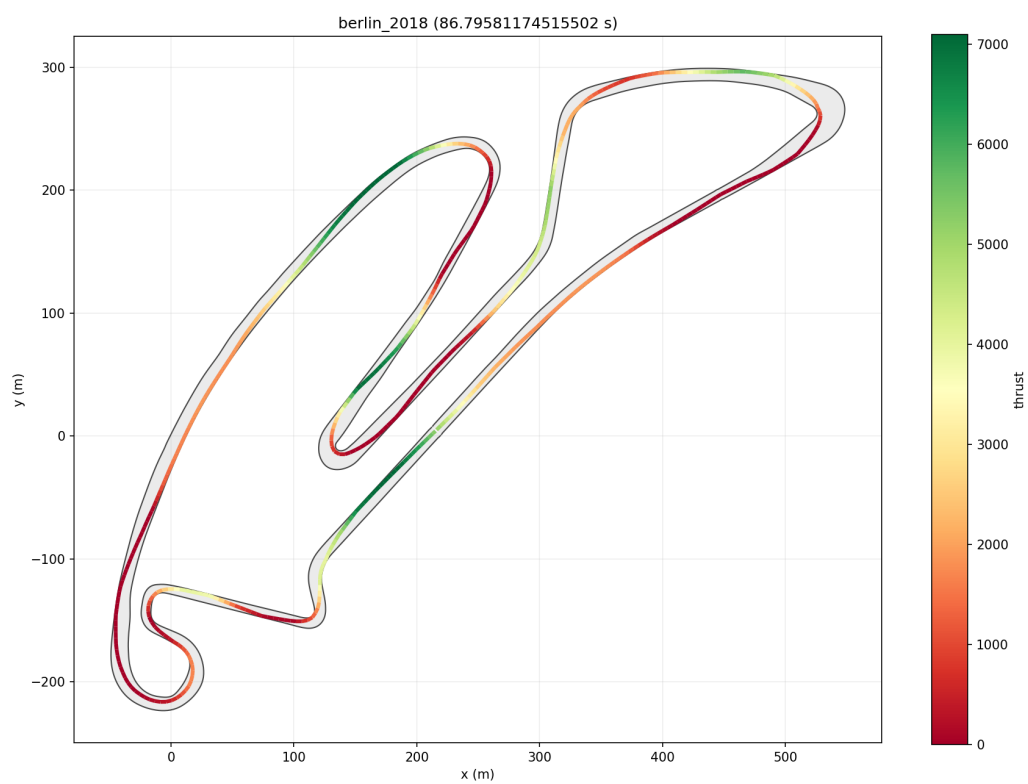


Fig. 5: Berlin 2018: longitudinal force profile along the track.

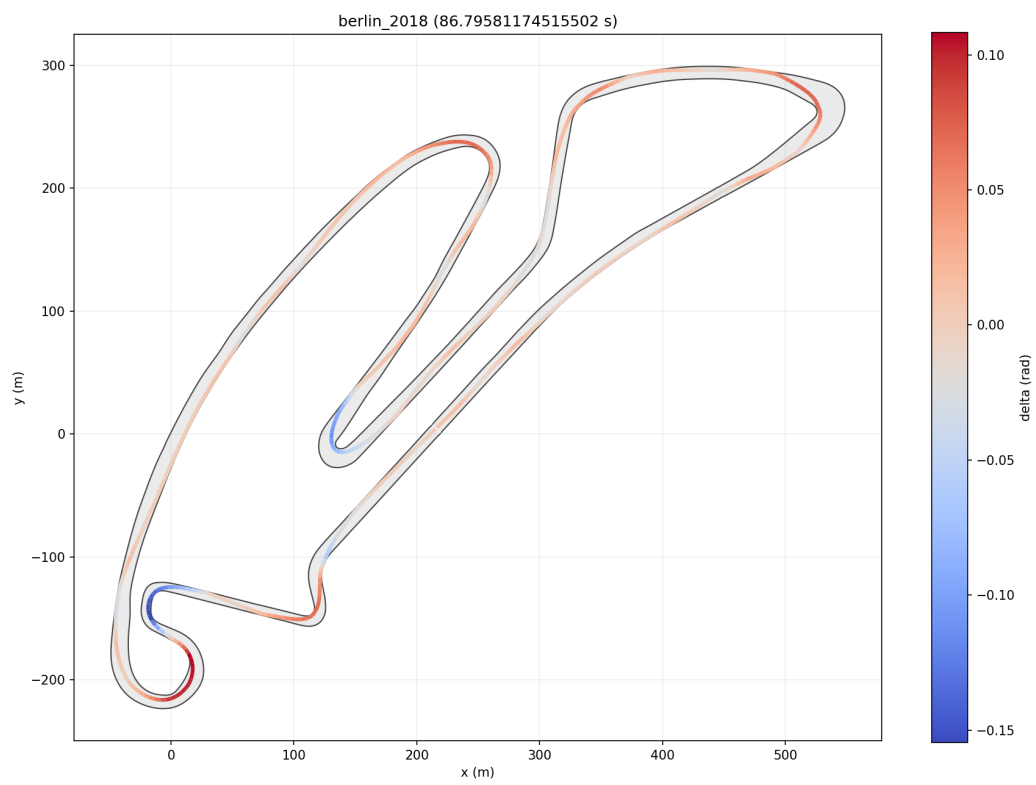


Fig. 6: Berlin 2018: steering angle evolution over the lap.

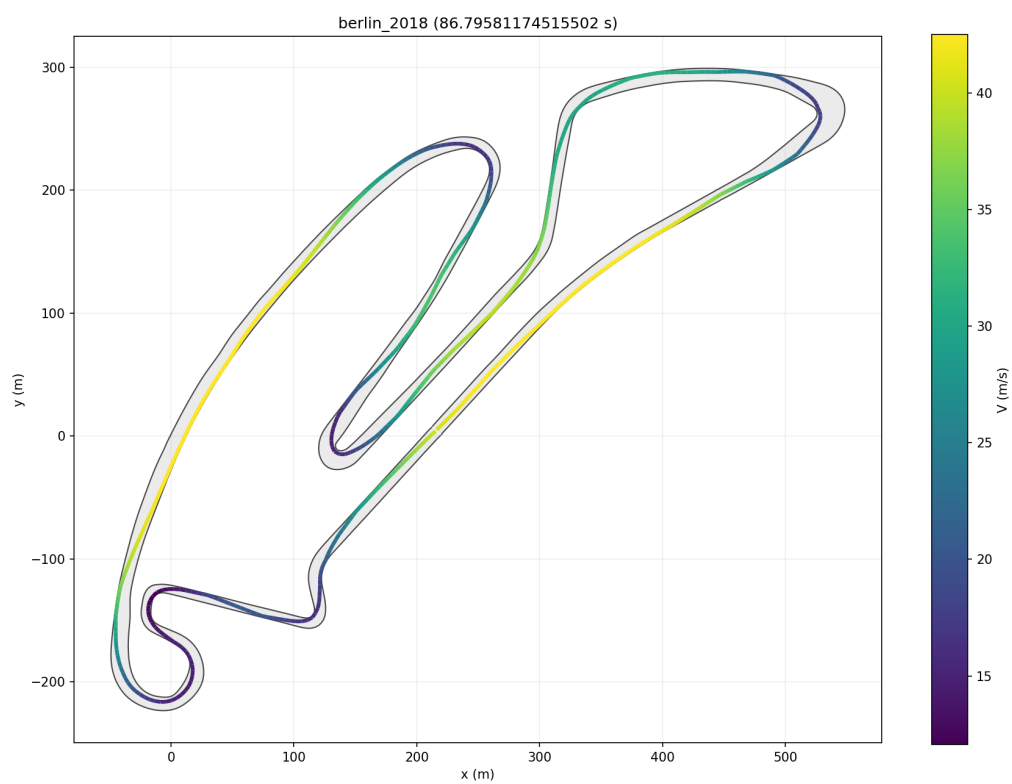


Fig. 7: Berlin 2018: vehicle velocity profile along the track.

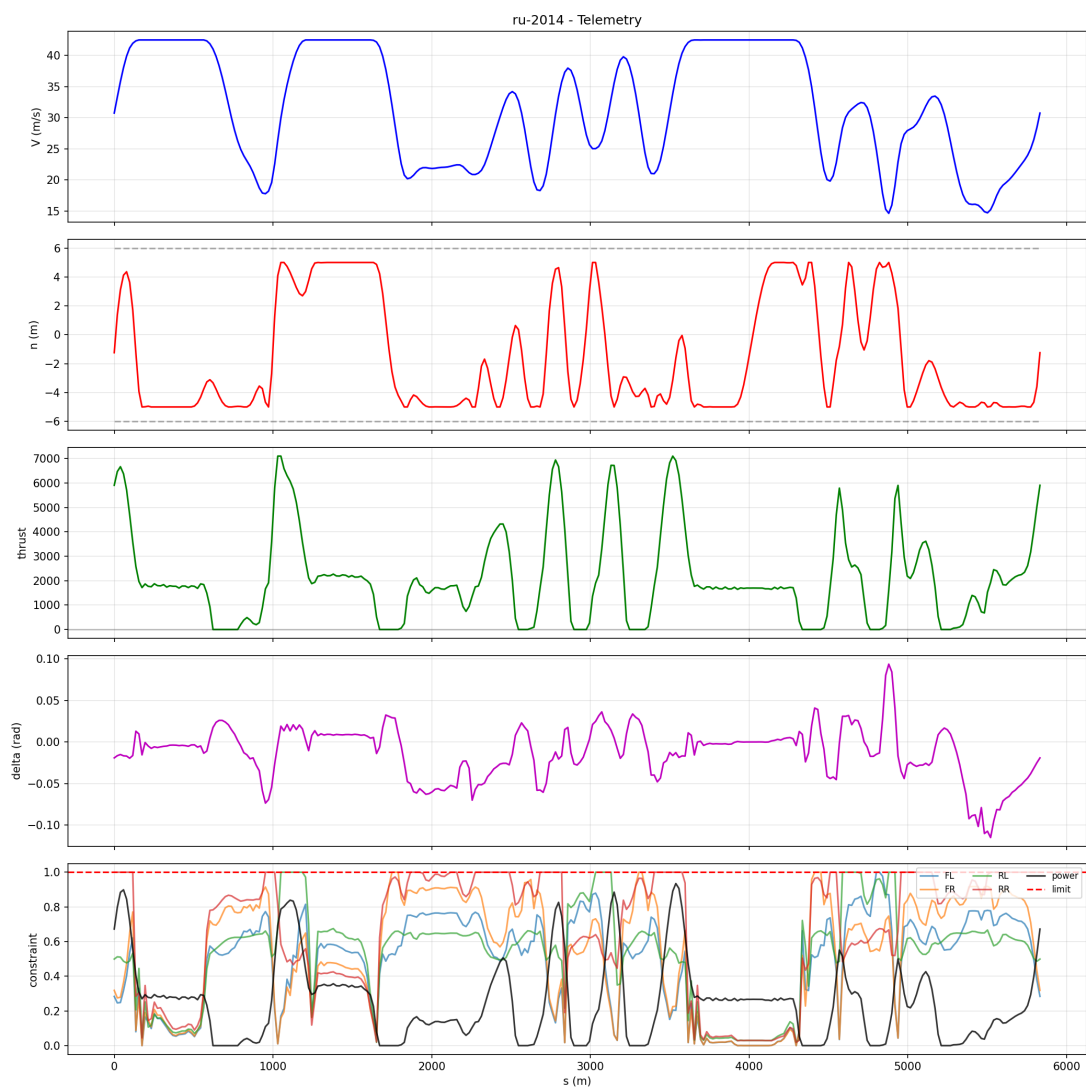


Fig. 8: Sochi 2014: vehicle telemetry over the lap.

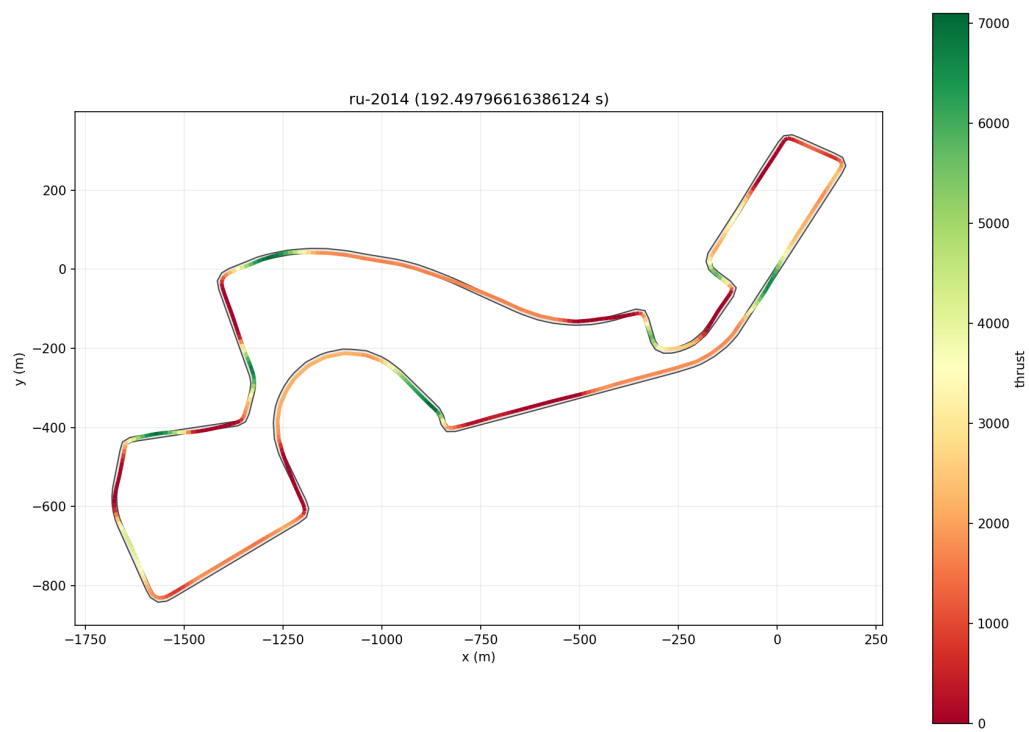


Fig. 9: Sochi 2014: longitudinal force profile along the track.

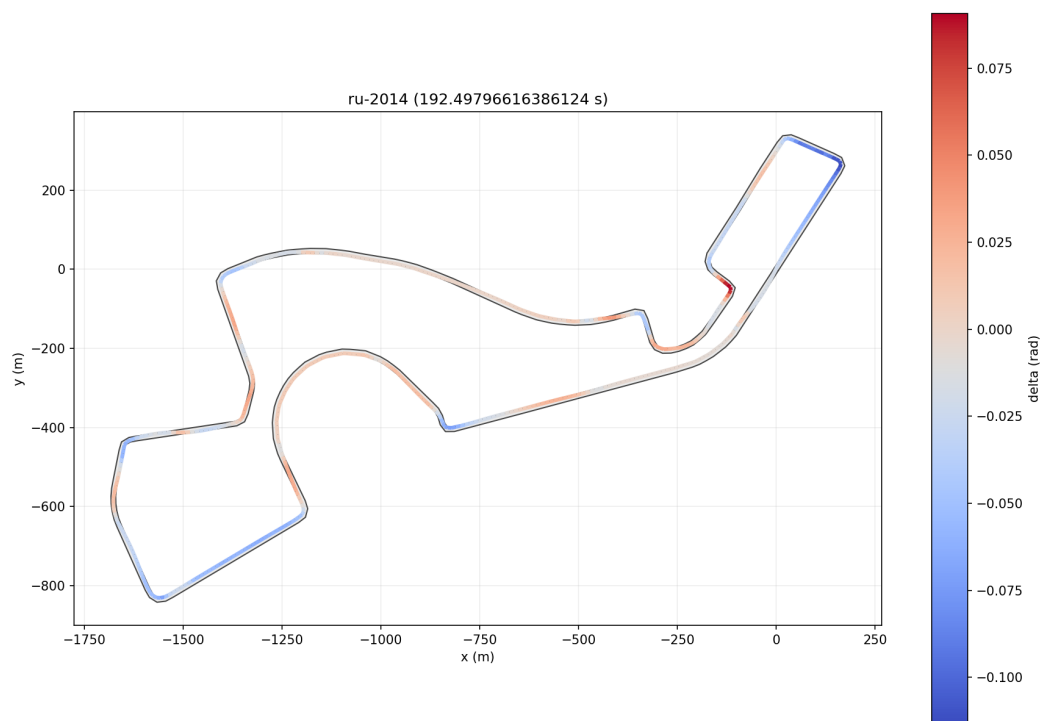


Fig. 10: Sochi 2014: steering angle evolution over the lap.

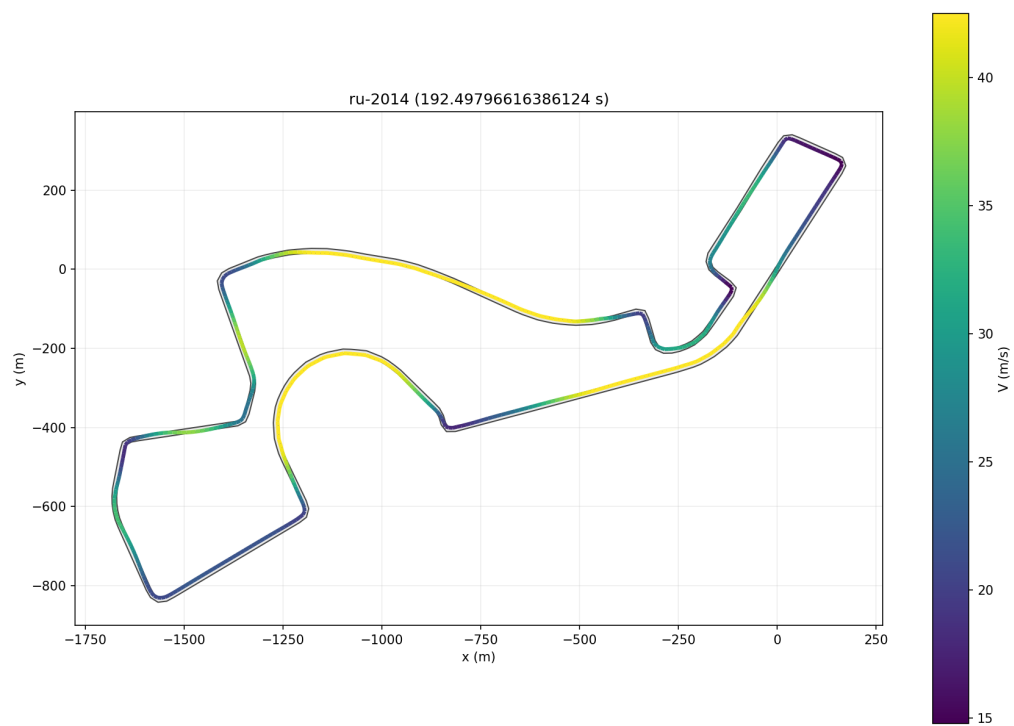


Fig. 11: Sochi 2014: vehicle velocity profile along the track.